FIFA 21 – Player Analysis and Recruitment Strategy Report

Power BI · R · DAX Project · FIFA 21 Data

Summary

The report moves from the overall player market (Page 1), to a comparison of clubs (Page 2), an in-depth analysis of the target club 1. FC Heidenheim 1846 (Page 3), and concrete actions: improving one existing player and recruiting for the missing position (Page 4). R is used to clean the data and build the regression model, DAX (Power BI) creates the grouping, ranking and scoring logic, and Power BI presents the results.

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Part I. Analysis Approach

1. Business questions

A club wants to raise the quality of its squad without a concrete budget plan. The report answers four consecutive questions:

- What does the player market look like (position, age, value, nationality)?
- Among the 681 clubs, which ones deserve a deeper analysis?
- Where is the selected club's squad strong and weak?
- What should be done: whom to develop, whom to recruit?

2. Data processing workflow

1. **Raw data:** the file "Data Cleaning Exercise fifa21_raw_data-21" with 18,979 rows × 77 columns.
2. **Cleaning with R:**
   - Remove stray characters and line breaks in text columns.
   - Convert Height (feet'inch) to inches and Weight (lbs) to kg.
   - Convert Value, Wage and Release Clause from text (e.g. €67.5M) to numbers so they can be calculated.
   - Remove the ★ character in the W.F, SM and IR columns.
   - Convert Joined to a date type and add the years_at_club column.
   - Create dummy variables: Left (left-footed), Striker, Goalie.
3. **Remove goalkeepers (GK):** 2,075 goalkeepers are removed because their stats (GK Diving, GK Handling, GK Reflexes…) differ completely from outfield players, so they cannot be compared with the remaining position groups.
4. **Clean data:** the field_players table with 16,904 players × 83 columns (Club and Contract added by splitting the Team & Contract column).
5. **Power BI:** load field_players, add the Target Club table, write DAX and build 4 pages.

3. The four pages and their roles

Page	Role	Question answered
1. Overall Player & Market Analysis	Market overview	Which positions, ages and nationalities offer an advantage when recruiting?
2. Club Comparison	Club screening	Which clubs qualify for a deeper analysis?
3. Squad Team	Squad diagnosis	Where is the squad strong or weak, and which position is missing?
4. Analysis & Improvement	Action	Whom to improve, which skills, whom to recruit?

Table 1. *The four pages and their roles*

Part II. Data Details

1. Scale

Item	Quantity
Tables in the model	2 (field_players, Target Club)
Players	16,904 rows (16,903 unique IDs; ID 251698 appears in 2 rows)
Total columns	83 = 77 original + 4 created by R + 2 split columns (Club, Contract)
Clubs	681; 1,146 players without a club
Objects with DAX	52 = 38 measures + 13 calculated columns + 1 calculated table

Table 2. *Data scale*

2. The ten folders in field_players

Each folder serves one analytical purpose. DAX appears only in folders 04 to 10 (plus the No Club measure in folder 01).

Folder	Content	DAX objects
01. Player Information	17 descriptive columns (name, nationality, club, contract, position, preferred foot…) and the No Club measure	1
02. Player Attributes	16 numeric columns on physique and age (Age, Height, Weight, Strength, Stamina, Sprint Speed…)	0
03. Player Performance	45 numeric columns on skills and performance (X.OVA, POT, PAC, DEF, PHY, Interceptions, Standing Tackle, Sliding Tackle…)	0
04. Age & Position Groups	6 calculated grouping columns: Age Group, OVA Group, Position Category, POT Group, Value Group, Wage Group	6
05. KPI & Summary	4 measures: Average Age, Average OVA, Average POT, Player Count	4
06. Position Benchmarks	15 measures: median OVA, POT, Value, Wage of each position group in the target club; median by BP position; Position Rank	15
07. Classification	8 classification and Pass/Fail check measures, 1 OVA Level column	9
08. Recruitment Scoring	3 measures: Recruitment Filter, Recruitment Priority, Recruitment Priority Level	3
09. Financial Analysis	Value, wage, club budget, Potential Gap, Value per OVA, Wage per OVA, Young Player %	12
10. Squad Selection	The Show in Starting XI measure and the Target Club table	2

Table 3. *The ten folders in field_players*

3. The Target Club table

Target Club is the list of clubs used as a slicer. When a club is selected, every measure in folders 06 to 10 is recalculated for that club. This is what lets the model be reused for any club.

Part III. Page-by-Page Explanation

Page 1. Overall Player & Market Analysis

Purpose: view the whole market to decide which age groups, positions and nationalities to look at when searching for players. Filters: Nationality, Position Category, Position, Club. Cards: 163 nations, 17K players, 681 clubs, 1,146 players without a club.

Chart	What it shows	Highest	Lowest	Trend
Average OVA by BP	Current quality by position	CF (71.3)	RM (64.6)	Declines from attackers (CF, LW, RW) down to the remaining positions
Average OVA & POT by Age Group	Current quality and potential by age	OVA highest at 30–34 and 35+ (about 68–69); POT highest at 15–19 (about 74)	OVA lowest at 15–19 (about 58)	OVA rises with age, POT falls with age; the two lines meet around the 25–29 group
Average Value & Wage by BP	Which positions the market values highly	CF	Full-back group and RM	Falls sharply from CF, LW, RW to CM, then flattens across the remaining positions
Number of Players by Position	Supply by position	CB	CF	Declines steadily; CB and ST are plentiful, CF is very rare
Donut Age Group	Age structure	20–24 (38%)	35+ (2%)	Concentrated in 20–29 (70%)
Donut Position Group	Position group structure	Midfielder (43%)	Attack (19%)	Midfielders and defenders make up the majority
Map and treemap	Player sources by nationality	England leads, followed by Germany, Spain, France, Argentina, Brazil	Small countries	Concentrated in Europe and South America

Table 4. *Charts on Page 1*

Related charts:

- Average OVA, Average Value & Wage by BP and Number of Players by Position are read together to see which positions have good quality, low wages and plentiful supply (recruitment opportunities), and which are expensive and scarce (hard to recruit).
- Average OVA & POT by Age Group and the Age Group donut are read together to choose a strategy by age: 15–19 for long-term investment, 20–24 offers the most choices, 25–29 is the group the market values highly, 30+ for immediate contribution.

DAX objects used: Age Group and Position Category (two grouping columns), Player Count, Average OVA, Average POT, No Club.

Key insight: age determines the recruitment target; attackers are valued the highest even though defenders and midfielders are the majority.

Page 2. Club Comparison

Purpose: Page 2 is not for drawing conclusions but is a **screening** step. With 681 clubs, users (executives, scouts, researchers) need to position each club on several criteria to select clubs eligible for deeper analysis. Filters: POT Group, Position Category, Position, Club, Age Group. Cards: 681 clubs, average OVA 66, average POT 71, average age 25.

Chart	What it shows	Highest	Lowest	Trend
Club table by Age Group	Avg POT, Avg OVA, Avg Wage, Sum of Value of each club	Big clubs (Juventus, Manchester City, Real Madrid…)	Small clubs	Value and wages concentrated in the 25–34 age group
Squad Quality by Age Group	Squad quality by age	POT highest at 15–19, OVA highest at 30–34 and 35+	OVA lowest at 15–19	Same as Page 1: youngsters have potential, older players have current ability
Squad Age Distribution	Number of players by age group	20–24 (about 6.4K)	35+ (about 0.4K)	Declines with age
Scatter POT & OVA (legend: OVA Group)	Current quality and potential of each club	Big clubs in the high POT, high OVA corner	Most clubs sit densely in the middle zone	Positive correlation
Scatter Wage & Value (legend: Wage Group)	Financial strength	Real Madrid, Barcelona, Manchester United at the extreme high end	Most clubs in the low zone	Positive correlation, strong gap between rich and poor
Scatter Value & OVA (legend: POT Group)	Expensive or cheap relative to quality	Attacking positions with high value	Defenders with low value	Some points show good OVA at a low price
Donut Position Group	Position structure across clubs	CB (22%)	RW, RM	CB and ST are the most numerous
OVA Position Group Performance	OVA by Attack, Defender, Midfielder group	Attack (about 71)	RM (about 64–65)	Attack high, midfield low and stable

Table 5. *Charts on Page 2*

Related charts: the three scatter charts together form a club positioning map on three axes: quality (OVA, POT), cost (Wage, Value) and efficiency (Value relative to OVA). Reading a point's position gives information: the top-right corner is a strong club; high POT but low OVA is a club with many young players; high OVA but low wage and value is an efficient club; a point far from the crowd is a special club that deserves a closer look. The three grouping columns OVA Group, Wage Group and POT Group (DAX) turn the points into groups that can be filtered and compared.

Result: narrowed from 681 clubs to one target club, here 1. FC Heidenheim 1846, chosen for further analysis.

Page 3. Squad Team

Purpose: diagnose the squad of the selected club (Club slicer = 1. FC Heidenheim 1846, other slicers left at All). Cards: average POT 71, average OVA 67, Potential Gap 5. The page has 8 charts and tables.

Component	What it shows	Conditions, related DAX
1. Overall Players table	All players of the club: age, position, X.OVA, POT, wage, value	Position Category column (DAX) colours rows by group; Potential Gap column (DAX) = POT minus OVA
2. Average table by Position Category	Gaps in OVA, wage and value between Defender, Midfielder, Attack	Uses the Position Category grouping column
3. Bar chart Player Count by BP	Which positions are crowded and which are short	Player Count measure

Component	What it shows	Conditions, related DAX
4. X.OVA – POT scatter (size = Value, colour by Age Group)	Seen visually instead of as numbers: young players with high POT and low OVA; older players with high OVA; who deviates from the crowd	Age Group column (DAX)
5. Show in Starting XI table	Best starting lineup by OVA	Position Rank and Show in Starting XI (DAX), explained below
6–8. Three detail tables: Attack, Defender, Midfielder	Each player in each group with OVA, POT, Value, Wage to see who needs improvement	Filtered by Position Category

Table 6. Components of Page 3

Highest, lowest and trends:

- Defenders have the highest average OVA (about 68), above Attack and Midfielder (about 66). Defender is therefore chosen as the base group to analyse first.
- By count, ST is the largest (5 players), CB and RM have 4, CAM, LM and LW have only 1, and there is no RW at all.
- On the scatter chart, young players sit high on the POT axis but low on OVA, and older players shift to the right.

How the Show in Starting XI table works

1. **Position Rank** ranks players of the same BP position in the selected club by OVA from high to low. If two players have the same OVA, the one with the smaller ID ranks first, so no two players share a rank.
2. **Show in Starting XI** sets the number of players per position: CB takes 2; LB, RB, CDM, CM, CAM, LW, ST and RW take 1 each. A player whose rank is less than or equal to the required number is flagged 1 and appears in the table.

Why only 9 of 11 positions appear:

- The goalkeeper does not appear because R removed GKs from the data from the start. DAX therefore selects only 10 outfield players.
- RW does not appear because the club has no player whose best position is RW, so nobody is ranked in this position.

Conclusion of Page 3: Heidenheim lacks a right winger (RW), which is the position to recruit. In addition, players with low OVA in the three detail tables are the ones to improve, and the Defender group is used as the benchmark to analyse first.

Page 4. Analysis & Improvement

Page 4 solves the two priority problems drawn from Page 3.

Block 1. Improving the OVA of O. Hüsing (centre-back, CB)

- The first three tables list players of the three groups Attack, Midfielder and Defender. Defender has the highest OVA, so it is analysed first.
- The fourth table lists Defender players with age, OVA, POT, PAC, DEF and PHY; the colour intensity shows high or low values for quick comparison.
- The fifth table expands the skills needed by defenders (Interceptions, Standing Tackle, Sliding Tackle, Heading Accuracy, Strength, Sprint Speed) for two players: P. Mainka (the benchmark) and O. Hüsing (to be improved).

- The analysis uses R: the OVA prediction model shows that Hüsing should prioritise Interceptions (about 13 points below the benchmark, largest regression coefficient), then Standing Tackle (about 6 points) and Sliding Tackle (about 5 points). A simulation of three scenarios shows that if he reaches Mainka's level in these three skills, his predicted OVA rises from about 69.54 to 75.68, an increase of about 6.14 points.

Block 2. Recruiting a player for the RW position

One single table lists RW players, combining columns and measures newly created in DAX to rank recruitment priority. The table is flexible because the club has no concrete plan for wage and value budgets: every threshold is calculated from the club selected in Target Club, so when the plan changes, the list and ranking can be recalculated.

Display conditions:

- The Club column is filtered to Blank, i.e. players who belong to no club and can be signed immediately.
- Position RW (298 players).
- **Recruitment Filter:** keeps only players whose OVA is above the club's lowest median OVA. The purpose is to remove players who would not improve the squad.

How the benchmark is built: from the target club, the median of each group Attack, Midfielder and Defender is calculated for OVA, POT, Value and Wage. Each metric has 3 medians; call L the smallest median, M the middle median and H the largest median. The classifications below compare a candidate with these three marks, so Low, Medium and High are relative to the club itself, not to the whole of FIFA.

Three-level classification:

Column	Low	Medium	High	Exception
OVA Classification	Above L, below M	From M to below H	H or above	At or below L: left blank (does not meet the minimum standard)
POT Classification	Below M	From M to below H	H or above	Left blank if data is missing
Value Classification	At or below L	From L to M	From M to H	**Over Budget** if above H
Wage Classification	At or below L	From L to M	From M to H	**Over Budget** if above H

Table 7. *Three-level classification for OVA, POT, Value and Wage*

For OVA and POT, the higher the level the better. For Value and Wage, the lower the level the better, and Over Budget means exceeding the highest level the club currently spends across position groups.

Why Recruitment Priority exists (maximum 100 points): points add up over four parts, favouring players who are good, have potential and are cheap.

Criterion	How points are given	Max points
OVA	High 40, Medium 30, Low 20	40
POT	High 30, Medium 20, Low 10	30
Wage	Low 15, Medium 10, High 5, Over Budget 0	15
Value	Low 15, Medium 10, High 5, Over Budget 0	15

Table 8. *How Recruitment Priority is scored*

The weights show the design priority: player quality (OVA and POT) accounts for 70 points and cost (Wage and Value) for 30 points. Players who do not pass the Recruitment Filter receive no score.

Recruitment Priority Level: the score is converted into four priority levels as shown below.

Recruitment Priority Level	Score
Top Priority	80 and above
High Priority	65 and above
Medium Priority	50 and above
Low Priority	Below 50

Table 9. *Recruitment Priority Levels*

Illustrative example (V. Radić in the table): OVA High (40) + POT Low (10) + Wage Low (15) + Value Medium (10) = 75 points, so he falls in the High Priority group. He ranks high thanks to good quality and low cost, even though his potential is not high.

Remark: in the visible part of the list, the highest score is about 75 and nobody reaches Top Priority. A plausible reason is that players with high OVA and POT and also low wage and value are very rare, because high quality usually comes with high cost. Good players who are Over Budget are downgraded even though their OVA is high.

Part IV. Summary, Status and Limitations

1. Summary of tools and results

Page	Main tools	Result
1	Grouping columns in DAX, KPI	Age and position determine the recruitment strategy
2	OVA Group, Wage Group, POT Group	Target club selected from 681 clubs
3	Position Rank, Show in Starting XI, Potential Gap	Missing RW identified; Defender chosen as the base group
4	R (regression) and the DAX classification and scoring chain	Development plan for O.Hüsing and a prioritized RW recruitment list

Table 10. *Summary of tools and results by page*

2. Current status

The report is still in progress. Next steps:

- Complete the OVA improvement analysis for the other positions.
- Prepare a reserve squad for players with high POT but still low OVA.

3. Limitations

- **Budget:** the club's budget policy (wage and value limits) must be known to select the most suitable players. The Over Budget thresholds are currently calculated from the target club's medians, so the recruitment list is for reference only until the real budget is available.
- **Scope:** so far the analysis covers one target club (1. FC Heidenheim 1846), one recruitment position (RW) and one player to improve (O. Hüsing).

4. Accompanying files

The source code is kept separately: one R file (data cleaning and the regression model) and one DAX file (measures, calculated columns and calculated table).